

Jianbo Liu^{1,*}
 Tianjian Zhang¹
 Hang Su²
 Xin Xu¹

Determination of Solubility and Metastable Zone of Sodium Hypophosphite and Nucleation Kinetics

To obtain thermodynamic and kinetic data of sodium hypophosphite, such as solubility and metastable zone, respectively, the solubility data for sodium hypophosphite in the temperature range of 298.15–373.15 K were obtained using a dynamic method. These data were then fitted to the Apelblat and Van't Hoff equations, and the corresponding model parameters were determined. The effects of stirring intensity and cooling rate on the width of the metastable zone of sodium hypophosphite in water were studied. The findings indicated that an increase in stirring intensity reduced the width of the metastable zone, whereas an increase in cooling rate resulted in its widening. The self-consistent Nývlt and Sangwal metastable zone models were employed to calculate nucleation dynamic parameters in conjunction with classical nucleation theory. The results showed that the nucleation order was 2.311–3.361 over the investigated temperature range. 316.15 K is the critical temperature point at which sodium hypophosphite transforms into a dominant nucleation mode. The solid–liquid interface tension decreased rapidly with the increase of saturation temperature, and the solid–liquid interface tension is 1.167–2.638 mJ m^{−2}.

Keywords: Crystal nucleation, Metastable zone, Sodium hypophosphite, Solubility

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1 Introduction

Sodium hypophosphite is an ideal reducing agent, which is mainly used in food, electroless plating, medicine, organic synthesis, and other industries [1]. According to the solubility characteristics of sodium hypophosphite, the batch cooling method is commonly used for production. The equipment conditions of the batch crystallization process are simple, but there are problems such as small unit processing capacity, large floor space, and high operating costs [2]. In contrast, the continuous crystallization process is a more efficient crystallization method because it can overcome the disadvantages of large product variation and low production efficiency of batch crystalline crystals. According to the analysis of Schaber et al. [3], the production cost can be reduced by about 9–40 % using continuous crystallization. Therefore, continuous crystallization is attracting more and more attention [4, 5].

To avoid burst nucleation caused by excessive supersaturation in the industrial crystallization process, the range of the metastable zone needs to be determined according to different operating conditions. The metastable zone is affected by many factors, including the cooling rate, impurity content, and solvent composition [6–9]. However, there is still a lack of basic data such as thermodynamics and kinetics of sodium hypophosphite crystallization. There are only solubility data for sodium

hypophosphite at 25 °C and 100 °C in the published literature [10]. By measuring the complete solubility data, metastable zone, and kinetic parameters of sodium hypophosphite, the nucleation and growth process can be further mastered, which can be used to predict the crystal size and distribution, control the crystal quality index, and assist in the design of the continuous industrial crystallization process of sodium hypophosphite.

To obtain the basic data of thermodynamics and kinetics of sodium hypophosphite crystallization, the solubility of sodium hypophosphite was determined using focused beam reflectance measurement (FBRM) combined with the dynamic method, and the solubility data of sodium hypophosphite were fitted using the Apelblat equation and the Van't Hoff equation. The effects of stirring intensity and cooling rate on the width of the metastable zone of sodium hypophosphite solution were also studied. The width of the metastable zone was predicted

¹Jianbo Liu (qingzhoujianbo@126.com), Tianjian Zhang, Xin Xu
 College of Engineering and Technology, Tianjin Agricultural
 University, 22 Jinjing Road, Xiqing District, Tianjin, P. R. China.

²Hang Su
 Tianjin Economic and Technological Development Area, Tianjin
 University of Science and Technology, 9 13th Avenue, Tianjin,
 P. R. China.

Table 1. The purity and suppliers of chemicals used in this study.

Chemical name	CAS number	Grade	Initial purity	Purification method	Final purity	Analytical method
NaPO ₂ ·H ₂ O	10039-56-2	AR	≥0.99	Recrystallization	≥0.995	Gravimetric method for hypophosphate

by using the self-consistent Nývlt equation and Sangwal nucleation theory, and the nucleation parameters, solid–liquid surface tension, frequency factor, and surface energy factor were calculated. These findings were a fundamental theoretical foundation for advancing sodium hypophosphite's continuous cooling crystallization process.

2 Materials and Methods

2.1 Reagents and Instruments

Reagent: Sodium hypophosphite monohydrate (analytical grade) was obtained from Chinese Medicine Chemistry Reagent Limited Corporation), whereas deionized water was produced in-house (at room temperature, with a pH of approximately 6.6 and electrical conductivity less than $1 \times 10^{-5} \text{ S m}^{-1}$). The purity and suppliers of chemicals used in this study are shown in Tab. 1.

Instrumentation used: Optimax™ 1001 Auto Chemical Synthetic Reactor from Metretoido; FBRM G400 Focus Beam Reflectance Measurement Device from Metreydo; SPR200 peristaltic Pump from Metrelito; XPR2004SC Electronic Balance from Metretiido.

2.2 Experimental Setup

The schematic diagram of the device for measuring the solubility and metastable zone data of sodium hypophosphite is shown in Fig. 1. The Optimax™ 1001 automatic chemical synthesis reactor had a volume capacity of 1 L. It was equipped with a highly accurate solid-state heat transfer module, allowing temperature control with an accuracy of $\pm 0.01 \text{ }^{\circ}\text{C}$. The fundamental prin-

ciple of FBRM involved employing a focused laser beam that rotated at a constant speed to scan the area between the probe window and the target being measured. When the focused laser beam sweeps through the crystal, the laser beam was reflected to FBRM and converted into the chord length of the crystal particles, to achieve the purpose of online detection of particle number and distribution [11]. The effective detection range of FBRM is 0.2–2000 μm . The solid phase is manually fed through a solid phase injection tube, and the liquid phase is stored in a beaker. It was connected to the Optimax™ liquid phase feed line via a peristaltic pump. Electronic balances, peristaltic pumps, temperature probes, and FBRM were all connected to the data logging and control terminal to monitor and record changes in key parameters:

1. electronic balance; 2. peristaltic pump; 3. FBRM; 4. liquid-phase sampling tube; 5. mechanical stirring; 6. solid-phase sampling tube; 7. temperature probe; 8. Optimax™ 1001 auto chemical synthetic reactor; 9. recording and control terminal.

2.3 Solubility Measurement Method

In this study, the solubility data of sodium hypophosphite were determined using the dynamic method [12, 13]. In addition, FBRM was used to detect the dissolution process of crystals in solution during solubility determination to ensure the accuracy of the data. At the beginning of the experiment, the solute and solvent were accurately weighed on an analytical balance with an accuracy of 0.1 mg and dissolved in a beaker (the solvent volume is 200 mL). The material was dosed through a peristaltic pump, and the FBRM system was turned on to detect the dissolution of solids in the solution. Then add 1–2 mg of solute after accurate weighing. During solute dissolution, the FBRM particle count curve gradually decreased to the particle number level of the clear solution. Repeat the process until the particle number curve no longer returns to the clear solution and the dissolution endpoint is reached, as shown in Fig. 2. Record the amount of all solutes and solvents added at this time. Solubility was determined at all temperature points according to the above operations. To ensure the accuracy of the experiment, the solubility determination experiment was repeated three times, and the arithmetic average was taken as the solubility value at this temperature.

The solubility for sodium hypophosphite was calculated as follows (Eq. 1):

$$x = \frac{m_1/M_1}{m_1/M_1 + m_2/M_2} \quad (1)$$

where x is the solubility of sodium hypophosphite, mol mol^{-1} ; m_1 and m_2 are the mass of solute and solvent, respectively, g; Additionally, M_1 and M_2 are the relative molecular masses of the solute and solvent, respectively, g mol^{-1} .

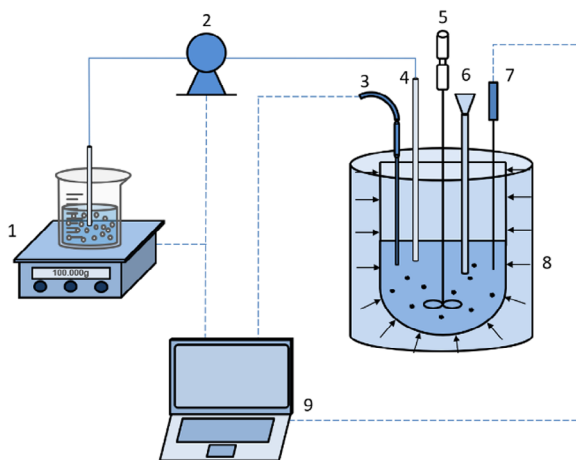


Figure 1. Flow chart of the crystallization experimental device.

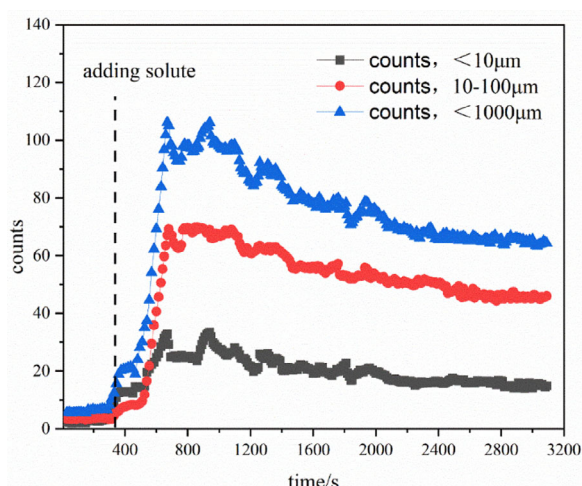


Figure 2. Curve depicting particle count data obtained from the focused beam reflectance measurement (FBRM) device.

2.4 Methodology to Determine the Metastable Zone

According to the solubility data obtained by the experiment, the sodium hypophosphite saturated solution at a certain temperature was accurately configured, and the Optimax end temperature was set to be 5–10 °C higher than the saturation temperature to ensure the complete dissolution of sodium hypophosphite crystals. Turn on the FBRM system, and reduce the temperature of the solution to saturation at a cooling rate of no more than 0.5 K min^{−1}. After the temperature was constant, set a certain stirring rate and cooling rate. When FBRM detected a sharp increase in the number of particles, the temperature at that time was recorded. To ensure the accuracy of the experiment, the metastable zone determination experiment was repeated three times, and the average value of the arithmetic was taken as the solubility value at this temperature.

2.5 Assessing the Accuracy of the Experimental Device

A preliminary examination of the solubility in the sodium hypophosphite–water system at 298.15 and 373.15 K was conducted to evaluate the reliability of the device. The obtained results were then compared with values from the existing literature, as illustrated in Tab. 2. As depicted in Tab. 2, it was evident that the experimental solubility data closely align with the values reported in the literature, with a relative error within the range of 0.3 %. This suggested that the method and apparatus employed

Table 2. The solubility data of sodium hypophosphite in water.

T/K	$x_{\text{ref}}/\text{mol mol}^{-1}$ [10]	$x_{\text{exp}}/\text{mol mol}^{-1}$
298	0.1440	0.1451
373	0.3059	0.3067

for determining solubility and the metastable zone in this study were accurate and dependable.

3 Results and Discussion

3.1 Solubility Determination

To simplify the calculations, the Apelblat model and the Van't Hoff model [14, 15] were used to fit the solubility data. Root mean square deviation (RMSD) was used to evaluate the difference between the measured solubility value and the model fit value. The RMSD was calculated using the following equation:

$$\text{RMSD} = \left[\sum_{i=1}^N \frac{(X_i^{\text{exp}} - X_i^{\text{cal}})^2}{N} \right]^{1/2} \quad (2)$$

where X_i^{exp} represents the experimental solubility data; X_i^{cal} represents the solubility data calculated by the Apelblat equation and the Van't Hoff equation; N represents the total number of measured solubility data.

3.1.1 Van't Hoff Model

Solubility can be described by the relationship between the logarithmic value of the molar fraction of solute and the absolute temperature due to the influence of the solvent:

$$\ln x = -\frac{\Delta H_s}{R_g T} + \frac{\Delta S}{R_g} \quad (3)$$

where x represents the solute solubility, mol mol^{−1}. T represents the saturation temperature, K. ΔH_s represents the enthalpy of solute dissolution, J mol^{−1}; ΔS represents the solute dissolution entropy, J mol^{−1} K^{−1}; R_g represents the gas constant, 8.314 J mol^{−1} K^{−1}.

3.1.2 Apelblat Model

The Apelblat equation was based on the Clausius–Crapeyron equation by assuming that the enthalpy of a solution becomes a curve of temperature. The solubility equation was derived as

$$\ln x = A + \frac{B}{T} + C \ln T \quad (4)$$

where x is the solute solubility, mol mol^{−1}; T is the saturation temperature, K; A , B , and C are the parameters of the equation.

3.1.3 The Fitting of Solubility Data

The correlation between the Apelblat equation and Van't Hoff equation was established using Eqs. (3) and (4), as illustrated in Fig. 3. This allowed for determining relevant parameters A , B , and C and the correlation coefficient R^2 . Tab. 3 presents the parameter values obtained through fitting these equations to the solubility data of sodium hypophosphite. Tab. 6 comprehensively summarizes the complete solubility data for sodium hypophosphite in water, encompassing the calculated values and the corresponding relative errors for various models. Notably, the

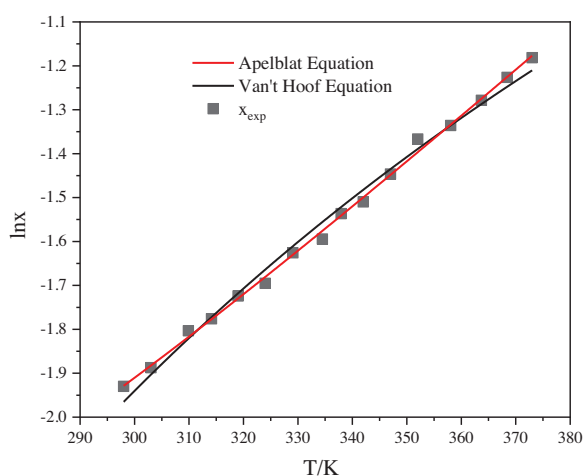


Figure 3. Modeling sodium hypophosphite solubility experimental data with Apelblat and Van't Hoff equations.

Table 3. The parameter values obtained by fitting sodium hypophosphite's solubility in water using the Apelblat and Van't Hoff equations.

Van't Hoff equation				Apelblat equation				
ΔH_S (J mol ⁻¹)	ΔS (J mol ⁻¹ K ⁻¹)	R^2	RMSD	A	B	C	R^2	RMSD
9300.85	14.87	99.54	0.298	-57.78	1795.89	8.75	99.74	0.460

Abbreviation: RMSD, root mean square deviation.

average relative errors for the Apelblat and Van't Hoff equations were 1.18 % and 1.85 %, respectively.

The correlation coefficient and RMSD values are presented in Tab. 3, and the relative deviation data are in Tab. 4. It indicated that both models exhibit minimal fitting errors. The RMS deviation was less than 0.5 %, demonstrating their effectiveness in accurately representing the solubility of sodium hypophosphite. This precision makes these models valuable for predicting solubility data at various temperature points within the system, offering essential data support for developing a continuous crystallization process for sodium hypophosphite.

3.2 Determination of Metastable Zone

3.2.1 Influence of Stirring Intensity on the Width of Metastable Zone

Ensuring alignment between crystal growth and nucleation models and the operational environment in industrial crystallizers for enhanced design and scaling of complex

mixing processes in industrial crystallization was important. The hydrodynamic properties of the solid-liquid phase were the basis for the scale-up of the device, which required the mixing Reynolds number ($N'Re$) and Froude number ($N'Fr$) to be constant in the model and industrial plant [16]. Therefore, in the process of industrial crystallization design, $N'Re$ and $N'Fr$ were mostly used as the basis for design scale-up. The industrial crystallization process does not have a strong vortex current, and the relationship between the Froude criterion and the metastable region is not considered in this study [17]. The turbulence intensity of the crystallization process had a significant effect on the metastable zone. Therefore, the study of the variation of the width of the metastable zone with the stirring Reynolds number ($N'Re$) could provide an important basis for the design and scale-up of the crystallizer [18]. The experimental device stirring paddle is shown in Fig. 4. The Optimax™ 1001 impeller was an open-blade turbine agitator that was typical of radial flow impellers. The number of blades was 4, the diameter of the blade was 45 mm, the width of the blade was 16 mm, the ratio of the diameter of the blade to the inner diameter of the groove was 1/3, and the angle between the blade and the horizontal plane was 45°. The formula for calculating the stirring Reynolds number is given in the following equation:

$$N'_{Re} = \frac{d_j^2 n \rho}{\mu} \quad (5)$$

Table 4. Measured values of sodium phosphite solubility and calculated values of the model.

T/K	$x_{exp}/\text{mol mol}^{-1}$	SD/%	Apelblat equation		Van't Hoff equation	
			$x_{cal}^A/\text{mol mol}^{-1}$	RD/%	$x_{cal}^V/\text{mol mol}^{-1}$	RD/%
298.15	0.1440	0.053	0.1462	1.53	0.1401	2.71
303.18	0.1515	0.07	0.1534	1.25	0.1494	1.39
310.05	0.1648	0.015	0.1636	0.73	0.1621	1.64
314.29	0.1693	0.02	0.1705	0.71	0.1702	0.53
319.17	0.1783	0.06	0.1788	0.28	0.1797	0.79
324.15	0.1836	0.064	0.1877	2.23	0.1896	3.27
329.23	0.1968	0.018	0.1975	0.36	0.2001	1.68
334.65	0.2030	0.015	0.2086	2.76	0.2114	4.14
338.09	0.2151	0.075	0.2160	0.42	0.2187	1.67
342.15	0.2210	0.036	0.2251	1.86	0.2274	2.90
347.15	0.2354	0.061	0.2370	0.68	0.2385	1.32
352.15	0.2549	0.027	0.2495	2.12	0.2495	2.12
358.21	0.2629	0.05	0.2658	1.10	0.2634	0.19
363.84	0.2784	0.04	0.2819	1.26	0.2764	0.72
368.57	0.2934	0.012	0.2962	0.95	0.2875	2.01
373.15	0.3059	0.022	0.3080	0.69	0.2981	2.55

Abbreviations: SD: standard deviation; RD: relative deviation.

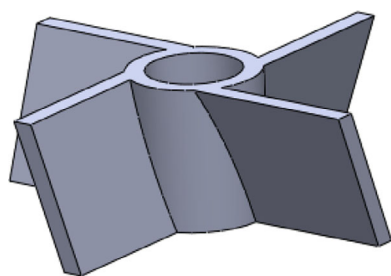


Figure 4. Optimax™ 1001 open-blade turbine impeller.

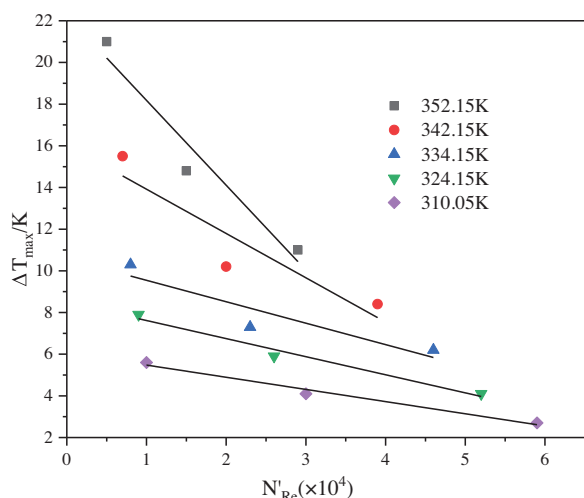


Figure 5. Influence of the stirring Reynolds number on the width of the metastable zone in a sodium hypophosphite solution.

where N'_{Re} is the stirring Reynolds number; d_j^2 is the agitator diameter, mm; n is the speed of the agitator, r/s ; ρ is the solution density, $kg\ m^{-3}$; μ is the dynamic viscosity of the solution, Pa.

The cooling rate was constant at $12\ K\ h^{-1}$, and the width of the metastable zone of the next sodium phosphate in water with different stirred Reynolds criteria was determined, as shown in Fig. 5. The width of the metastable zone of sodium hypophosphite in water narrowed with the increase of the stirring Reynolds number when the saturation temperature was constant. This was an increase in turbulence intensity that results in an increase in the frequency of collisions between the crystals in the solution and the wall and the impeller, thereby accelerating the rate of secondary nucleation. The effect of saturation temperature on the width of the metastable zone was more significant, as shown in Fig. 6. The metastable zone becomes widened as the saturation temperature decreases. This was because with the increase of the viscosity of the solution, the relative flow velocity of solute molecules and tiny particles decreased, the probability of collision was greatly reduced, and the formation of stable crystal nuclei was more difficult.

3.2.2 Effect of Cooling Rate on Metastable Zone Width

The widths of the metastable zone of sodium hypophosphite at different saturation temperatures at 3, 6, 12, and $24\ K\ h^{-1}$ were

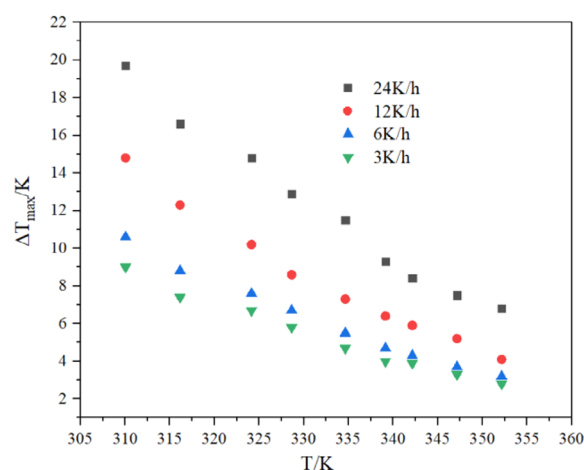


Figure 6. Influence of the cooling rate on the metastable zone of sodium hypophosphite.

determined at $300\ r\ min^{-1}$, as shown in Fig. 6. It could be clearly seen from the figure that the width of the metastable zone ΔT_{max} increased with the increase of the cooling rate. The effect of cooling rate on nucleation was mainly manifested in the increase rate of supersaturation of the solution, and the acceleration of the accumulation rate of supersaturation would increase the nucleation rate. However, according to the crystal growth theory [17], there would be repeated dissolution and growth processes before the tiny particles reach the critical nucleus size to form a stable crystal nucleus. Depending on the surface energy of the crystal and the generation energy, the time required to form a stable crystal nucleus varies greatly. A faster cooling rate would shorten the time to reach the same maximum subcooling degree. In a short period of time, the tiny particles were still in the process of forming stable crystal nuclei, resulting in the increase of the width of the metastable region.

3.3 Analysis of Crystallization Nucleation Mechanism of Sodium Hypophosphite

3.3.1 Self-Consistent Nývlt Metastable Zone Equation

The Nývlt metastable zone equation [19] expressed the relationship between the width of the metastable zone and the nucleation dynamics, which was used to predict the relationship between the width of the metastable zone ΔT_{max} and the cooling rate R . The expression was as follows:

$$\Delta T_{max} = \left(\frac{dC_0}{dT} \right)_T^{(1-m)/m} \left(\frac{R}{K} \right)^{1/m} \quad (6)$$

where m represents the nucleation orders; c_0 represents the molar fraction solubility of the solution, $mol\ mol^{-1}$; $(dc_0/dT)_T$ represents the slope of the solubility curve at temperature T ; K is the nucleation constant, which represents the number of critical nuclei per unit volume and per unit of time. Its size depends on the process by which the solute molecules form stable crystal

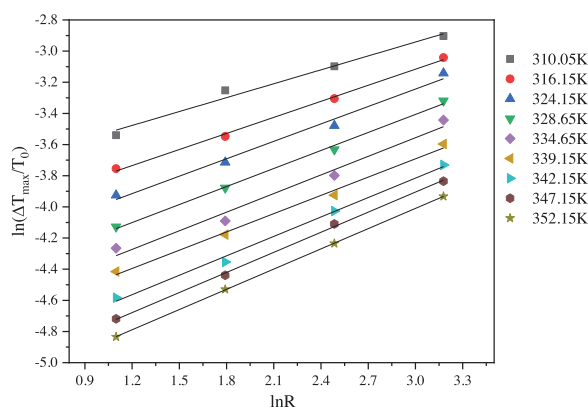


Figure 7. Relationship between $\ln(\Delta T_{\max}/T_0)$ and $\ln R$ at different saturation temperatures.

nuclei, as well as the precision of the method used to determine the width of the metastable zone.

By applying logarithms to both sides of Eq. (6), the following equation was obtained:

$$\ln(\Delta T_{\max}) = \frac{1-m}{m} \ln\left(\frac{dC_0}{dT}\right)_T - \frac{1}{m} \ln K + \frac{1}{m} \ln R \quad (7)$$

where the value of $(dC_0/dT)_T$ can be calculated using the solubility curve. Therefore, the values of m and K could be calculated from the linear relationship between $\ln(\Delta T_{\max})$ and $\ln R$.

Sangwal made a correction to the classical Nývlt metastable zone equation [20]. The ratio of the supercooling degree ΔT to the saturation temperature T_0 of the solution during the cooling process was correlated with the cooling rate, which was convenient for direct observation of the influence of saturation temperature on the width of the metastable zone:

$$\ln\left(\frac{\Delta T_{\max}}{T_0}\right) = \frac{1-m}{m} \ln\left(\frac{\Delta H_S}{R_G T}\right)_T + \frac{1}{m} \ln\left(\frac{f}{K T_0}\right) + \frac{1}{m} \ln R \quad (8)$$

where T_0 is the saturation temperature of the solution, K ; T is the temperature of the solution, K ; f is the proportionality constant, which represents the number of particles per unit volume; the value of f is controlled by the process of aggregation and diffusion in the solution, $f = 10^{27} \# \text{ m}^{-3}$.

Furthermore, $A = \frac{1-m}{m} \ln\left(\frac{\Delta H_S}{R_G T}\right)_T + \frac{1}{m} \ln\left(\frac{f}{K T_0}\right)$, $B = \frac{1}{m}$, and linear relationship of $\ln(\frac{\Delta T_{\max}}{T_0})$ with $\ln R$ could be obtained as fellows (Eq. 9):

$$\ln\left(\frac{\Delta T_{\max}}{T_0}\right) = A + B \ln R \quad (9)$$

The relationship between the metastable zone and the cooling rate was obtained by fitting the above formula, as shown in Fig. 7. The results of slope B , intercept A , nucleation orders m , nucleation constant K , and correlation coefficient R^2 are shown in Tab. 5.

It could be seen from Tab. 5, the nucleation orders m decreased with increasing saturation temperature. The nucleation order m represented the interaction between solute and solvent in solution; the smaller the m value, the more conducive to the

Table 5. Model parameter values calculated by the self-consistent Nývlt equation.

T_0/K	A	B	m	$K/10^{26}$	R^2
310.05	−3.834	0.298	3.361	1.009	0.9744
316.15	−4.147	0.344	2.910	2.651	0.9957
324.15	−4.362	0.373	2.682	4.450	0.9812
328.65	−4.563	0.386	2.593	7.162	0.9947
334.65	−4.750	0.398	2.511	10.748	0.9682
339.15	−4.865	0.391	2.557	14.693	0.9910
342.15	−5.062	0.416	2.406	20.558	0.9930
347.15	−5.195	0.430	2.326	25.319	0.9981
352.15	−5.308	0.433	2.311	31.853	0.9999

formation of a stable crystal nucleus, and the narrower the width of the metastable zone. According to the classical nucleation theory, the nucleation rate was closely related to the viscosity value of the solution. At lower temperatures, the viscosity value of the solution increased, resulting in an increase in the nucleation order, and the ability to form stable crystal nuclei was weaker [21]. When the saturation temperature was higher than 316.15 K, $m < 3$ means that the instantaneous nucleation of sodium hypophosphite in aqueous solution was the dominant nucleation mode. $m > 3$, the dominant mode of solution nucleation was continuous nucleation [22]. The nucleation constant K represented the number of critical nuclei per unit volume per unit time. The number of critical crystal nuclei formed at low temperatures was significantly reduced, and the nucleation ability was weaker [23].

3.3.2 Sangwal Metastable Zone Equation

The metastable zone width prediction model proposed by Sangwal could also explain the relationship between the metastable zone and nucleation dynamics. According to the classical three-dimensional nucleation theory [24], the nucleation rate of the homogeneous nucleation process was expressed as follows:

$$J = A_0 \exp\left[-\frac{16}{3} \pi \left(\frac{\gamma \Omega_V^{2/3}}{K_B T}\right)^3 / (\ln S)^2\right] \quad (10)$$

Eq. (10) relates the nucleation rate to temperature, supersaturation coefficient, and the physical properties of the crystalline substance. Here, J is the nucleation rate, $\#/(m^3 \text{ s})$; A_0 is the nucleation kinetics; γ is the solid–liquid interfacial tension, (mJ m^{-2}) ; Ω_V is the solute molecular volume, m^3 ; K_B is the Boltzmann constant, J K ; S is the supersaturation ratio.

According to the conventional solution theory, there was the following relationship between the temperature T and T_0 corresponding to the supersaturated concentration C and the saturated concentration C_0 :

$$S = \frac{C_0}{C_1} = \exp\left(\frac{\Delta H_S}{R_G T} \frac{\Delta T}{T_0}\right) \quad (11)$$

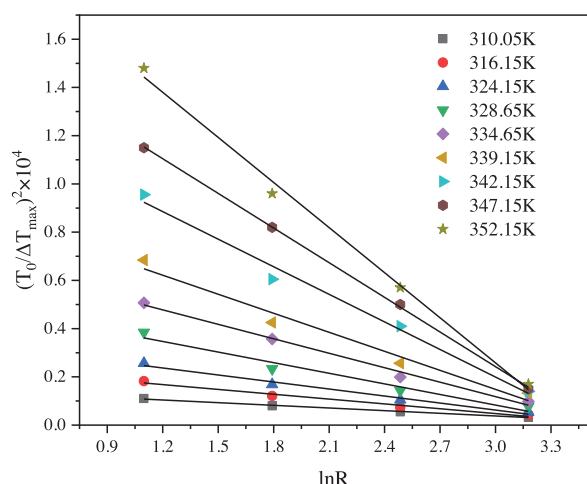


Figure 8. Relation between $(T_0/\Delta T_{\max})^2$ and $\ln R$ at different saturation temperatures.

The relationship between the nucleation rate near the boundary of the metastable zone and the rate of change of supersaturation was shown in the following equation:

$$J = f \left(\frac{\Delta H_s}{R_G T} \frac{R}{T_0} \right) \quad (12)$$

Combining Eqs. (11)–(13) and then taking the logarithm yields the following equation:

$$\left(\frac{T_0}{\Delta T_{\max}} \right)^2 = \frac{3}{16\pi} \left(\frac{\Delta H_s}{R_G T} \right)^2 \left(\frac{K_B T}{\gamma \Omega_V^{2/3}} \right)^3 \times \left[\ln \left(\frac{AR_G T}{f \Delta H_s} \right) + \ln T_0 - \ln R \right] \quad (13)$$

$$F_1 = \frac{3}{16\pi} \left(\frac{\Delta H_s}{R_G T} \right)^2 \left(\frac{K_B T}{\gamma \Omega_V^{2/3}} \right)^3, \quad E = \ln \left(\frac{AR_G T}{f \Delta H_s} \right),$$

$$F = F_1 (E + \ln T_0)$$

$$\left(\frac{T_0}{\Delta T_{\max}} \right)^2 = F - F_1 \ln R \quad (14)$$

Eq. (13) presents the linear relationship between two variables, denoted as $\left(\frac{T_0}{\Delta T_{\max}} \right)^2$ and $\ln R$. The correlation between the metastable zone of sodium hypophosphite and the cooling rate was modeled by Eq. (14), as depicted in Fig. 8. The saturation temperature had little effect on the nucleation kinetic factor A_0 but had a greater effect on the solid–liquid interface tension γ . γ indicated the difficulty level of solute crystallization in the solution. The smaller the γ , the less energy needed to be overcome to crystallize the solute [25, 26]. It could be seen from Tab. 6, the solid–liquid interface tension γ decreased with the increase of saturation temperature. It might be because the viscosity decreases of the solution due to the increase in temperature, resulting in a corresponding decrease in the solid–liquid interfacial tension γ . It was also consistent with the conclusion that critical crystal nuclei were more likely to form at medium and high temperatures in Section 3.3.1. The change of solid–liquid interface energy could be used as the basis or reference

Table 6. Model parameter values calculated by Sangwal's equation

T_0/K	F	F_1	E	$\gamma/(mJ\ m^{-2})$	$A_0/(10^{26})$	R^2
310.05	0.148	0.0368	−1.709	2.638	7.241	0.9921
316.15	0.248	0.0667	−2.039	2.206	5.206	0.9662
324.15	0.353	0.0970	−2.138	1.997	4.715	0.9774
328.65	0.523	0.147	−2.228	1.764	4.308	0.9481
334.65	0.718	0.200	−2.223	1.620	4.333	0.9876
339.15	0.937	0.263	−2.263	1.498	4.160	0.9588
342.15	1.343	0.381	−2.313	1.336	3.960	0.9817
347.15	1.679	0.479	−2.344	1.256	3.838	0.9995
352.15	2.128	0.623	−2.450	1.167	3.451	0.9927

for the selection of crystallization temperature in the industrial crystallization process. Lower crystallization temperatures mean that solute crystallization was more difficult. Therefore, in the continuous crystallization process, excessive reduction of crystallization temperature to improve crystal yield might cause the supersaturation of the system to be ununiform until many new crystal nuclei were initiated. This resulted in periodic oscillations in the particle size and distribution of crystal products [27].

4 Conclusion

The solubility of sodium hypophosphite in water was measured using the dynamic method. The results showed that the solubility of sodium hypophosphite increased with the increase of temperature, and the width of the metastable zone of sodium hypophosphite became narrow with the increase of stirring intensity. The solubility and metastable zone of sodium hypophosphite could provide a design basis for the selection of circulation capacity and supercooling degree of a continuous cooling crystallization system. Furthermore, the nucleation kinetic parameters m and nucleation constant K were calculated using the self-consistent Nývlt metastable zone equation and the experimental data of the metastable zone. It shows that the m value decreases with the increase of saturation temperature, and the K value increases with the increase of saturation temperature. 316.15 K is the critical point of sodium hypophosphite nucleation; above this temperature, the dominant nucleation mode is instantaneous nucleation; below this temperature, the dominant nucleation mode is continuous nucleation. The solid–liquid interface tension γ and nucleation kinetics factor A_0 were calculated by using the Sangwal metastable zone equation, which demonstrated that the saturation temperature has little effect on the nucleation kinetic factor A_0 but has a large γ on the solid–liquid interface tension, which decreases rapidly with the increase of saturation temperature. The study of nucleation dynamics holds crucial implications for guiding the design of crystallizers.

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Symbols used

A	[–]	Apelblat equation parameters
A_0	[–]	Nucleation kinetics factor
B	[–]	Apelblat equation parameters
C	[–]	Apelblat equation parameters
c_0	[mol mol ^{−1}]	The initial concentration of the solution
Δc	[mol mol ^{−1}]	Differential concentration
dj	[mm]	Mixing paddle diameter
f	[#/m ³]	Proportionality constant of Sangwal metastable zone equation
ΔH_s	[J mol ^{−1}]	Enthalpy of solute dissolution
J	[#/(m ³ s)]	Nucleation rate
K	[#/(m ³ s)]	Nucleation constant
K_B	[J K]	Boltzmann constant
M_1	[g mol ^{−1}]	Molecular weight of solute
M_2	[g mol ^{−1}]	Molecular weight of solvent
m	[–]	Nucleation orders
m_1	[g]	Solute mass
m_2	[g]	Solvent mass
N	[–]	Solubility measurement number of groups
$N're$	[–]	Stirring Reynolds number
n	[r/s]	Paddle speed
R	[K/h]	Cooling rate
RG	[J/(mol K)]	Gas constant
S	[–]	Relative supersaturation
ΔS	[J mol ^{−1} K ^{−1}]	Entropy of dissolution
T	[K]	Thermodynamic temperature
T_0	[K]	Solution saturation temperature
ΔT_{max}	[K]	Metastable zone width
x	[mol mol ^{−1}]	Sodium hypophosphite solubility
x_i^{exp}	[mol mol ^{−1}]	Experimental value of solubility of sodium hypophosphite
x_i^{cal}	[mol mol ^{−1}]	Calculated value of sodium hypophosphite solubility

Greek letters

γ	[m] m ^{−2}	Solid-liquid interfacial tension
μ	[Pa·s]	Dynamic viscosity of solution
ρ	[kg m ^{−3}]	Solution density
Ω_V	[m ³]	Molecular volume of sodium hypophosphite

Abbreviations

FBRM	focused beam reflectance measurement
RMSD	root mean square deviation
RD	relative deviation
SD	standard deviation

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